

Can We Treat Noisy Labels of Ambiguous Instances as Accurate?

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Abstract

Noisy labels significantly hinder the accuracy and generalization of machine learning models, particularly when resulting from ambiguous instance features that complicate correct labeling. Traditional approaches, such as those relying on transition matrices for label correction, often struggle to effectively resolve such ambiguity, due to their inability to capture complex relationships between instances and noisy labels. In this paper, we propose EchoAlign, a paradigm shift in learning from noisy labels. Unlike previous methods that attempt to correct labels, EchoAlign treats noisy labels ($\tilde{Y}$) as accurate and modifies corresponding instances (X) to better align with these labels. The EchoAlign framework comprises two main components: (1) EchoMod leverages controllable generative models to selectively modify instance features, achieving alignment with noisy labels while preserving intrinsic instance characteristics such as shape, texture, and semantic identity. (2) EchoSelect mitigates distribution shifts introduced by instance modifications by strategically retaining a substantial subset of original instances with correct labels. Specifically, EchoSelect exploits feature similarity distributions between original and modified instances to accurately distinguish between correctly and incorrectly labeled samples. Extensive experiments across three benchmark datasets demonstrate that EchoAlign significantly outperforms state-of-the-art methods, particularly in high-noise environments, achieving superior accuracy and robustness. Notably, under 30% instance-dependent noise, EchoSelect retains nearly twice the number of correctly labeled samples compared to previous

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methods, maintaining 99% selection accuracy, thereby clearly illustrating the effectiveness of EchoAlign. The implementation of EchoAlign is publicly available at <https://github.com/KevinCarpricorn/EchoAlign/tree/main>.

Keywords: Learning from Noisy Labels, Controllable Generative Models, Instance Modification, Feature Alignment, Sample Selection, Robust Machine Learning.

1. Introduction

The rapid advancement of neural networks has underscored the significance of learning from noisy labels (LNL) (Tan and Le, 2019; Dosovitskiy et al., 2021; Stiennon et al., 2020; Chen et al., 2023a). Although web crawling and crowdsourcing provide cost-effective means for collecting large datasets, they often introduce noisy labels that hinder model generalization (Yu et al., 2018b; Li et al., 2017; Welinder et al., 2010; Zhang et al., 2017; Natarajan et al., 2013; Gu et al., 2023). Recent studies have highlighted that label noise in pretraining data adversely affects the out-of-distribution generalization of foundation models in downstream tasks (Chen et al., 2023a, 2024). Noisy labels are generally categorized as random, class-dependent, or instance-dependent, with the latter two posing particular challenges due to ambiguous instance features, making it difficult to distinguish mislabeled examples from true class instances (Menon et al., 2018; Xia et al., 2020; Yao et al., 2023a; Bai et al., 2023).

Prior research has primarily approached LNL through either noise-modeling-free or noise-modeling frameworks. Noise-modeling-free techniques, such as filtering out high-loss examples (Han et al., 2018; Yu et al., 2019; Wang et al., 2019), are limited to selecting clean samples and do not address the potential for correcting incorrect labels, thereby discarding valuable supervisory information. In contrast, noise-modeling approaches explicitly consider the label-noise generation process (Scott et al., 2013; Scott, 2015; Goldberger and Ben-Reuven, 2016), often employing a transition matrix to relate noisy labels to their clean counterparts (Berthon et al., 2021). Theoretically, an optimal classifier can be trained with sufficient noisy data and an accurate transition matrix (Reed et al., 2014; Liu and Tao, 2015). However, estimating this matrix is inherently ill-posed due to uncertainty and variability in noisy data (Xia et al., 2019; Cheng et al., 2020). Moreover, these models often rely on additional assumptions, such as the exact nature of the noise, which are

challenging to validate and may not hold in real-world datasets, leading to suboptimal performance (Xia et al., 2020; Yao et al., 2023b; Liu et al., 2023). Traditional label correction methods are particularly limited when dealing with label noise caused by ambiguous features. For example, in datasets collected through web crawling, an image labeled as ‘dog’ might actually depict a cartoon or a product featuring a dog, which makes label correction challenging and often impractical.

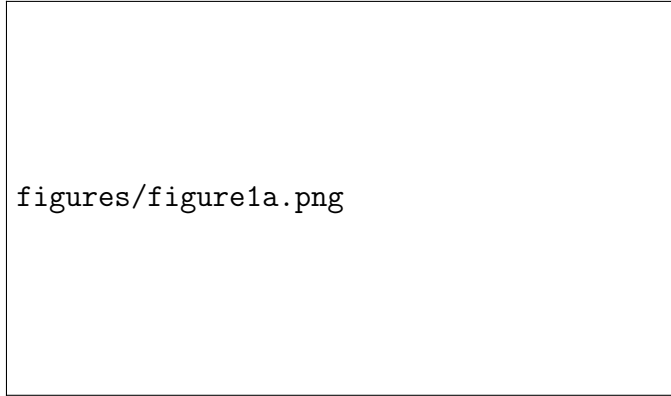


Figure 1: Instance modification effectively aligns instances with their labels, while label correction struggles with ambiguous cases.

In this paper, we introduce a novel perspective on handling label noise by employing *instance modification* rather than correcting labels. Instead of attempting to correct noisy labels, we adjust instances to better align with their labels, even if those labels are incorrect. This innovative approach, illustrated in Figure 1, directly addresses the root cause of label noise. Leveraging causal learning principles (Neuberg, 2003; Peters et al., 2017; Yao et al., 2021), we model instance-dependent label noise from a causal perspective, as depicted in the causal graph of Figure 3. Specifically, we consider how different factors, such as instance characteristics and latent variables, contribute to the generation of noisy labels, enabling us to better understand and address the root causes of label noise. In crowdsourcing scenarios, for instance, ambiguous or blurred instances are more prone to labeling errors. Instead of attempting to infer the ‘true’ label, modifying the instance to make it more distinguishable can be more effective. For example, in medical imaging, if a tumor is labeled as malignant but its visual features are too subtle, enhancing the image to highlight relevant features can assist both

models and humans in identifying it more accurately (Mirza et al., 2023). Similarly, in sentiment analysis, modifying ambiguous sentences to be more explicit can better align them with their intended sentiment labels, thereby reducing ambiguity and improving classification accuracy.

figures/figure1b.png

(a) Characteristic Shift.

figures/distribution_shift.png

(b) T-SNE Visualization of CIFAR-10 instance representations by using X .

figures/distribution_shift_fixed.png

(c) T-SNE Visualization of CIFAR-10 instance representations by using X'

Figure 2: (Top) Main challenge 1: Characteristic Shift. (Bottom) Main challenge 2: Distribution Shift.

However, instance modification presents challenges at both the instance and dataset levels. At the instance level, a key challenge is modifying instances while preserving their essential characteristics, such as shape, texture, or color patterns. These features are crucial for distinguishing related categories. Excessive alterations can distort these defining features, leading to a phenomenon we refer to as the characteristic shift (Figure 2a). For example, when a wolf is mislabeled as a dog, transforming the wolf image into a typical dog might eliminate important shared features such as body shape and texture, which are critical for distinguishing between wolves and dogs.

At the dataset level, instance modification may introduce a distribution shift (Figures 2b and 2c), where the statistical distribution of modified instances deviates from that of the original instances, potentially affecting model generalization to real-world data (Han et al., 2022a,b). Empirical observations, as visualized through T-SNE projections, reveal that modified instances may occupy a distinct feature space from their unaltered counterparts, altering the training dynamics. Addressing these shifts requires a balanced framework that preserves essential characteristics while effectively managing distribution differences between original and modified instances to ensure robustness in real-world applications.

To address these challenges, we propose a simple yet effective framework, EchoAlign (§4). EchoAlign consists of two key components: EchoMod and EchoSelect. EchoMod modifies instances using controllable generative models, ensuring alignment with noisy labels while preserving intrinsic characteristics. EchoSelect mitigates covariate shifts by selecting original instances with correct labels, maintaining a balanced distribution between original and modified data. This selection is guided by a novel insight: after instance modification, the cosine feature similarity between original and modified images reveals distinctions between correctly and incorrectly labeled samples. EchoSelect uses this similarity metric to curate a reliable training set, improving robustness and accuracy in both supervised and self-supervised training.

Our key contributions and findings are summarized as follows:

1. We introduce a transformative shift in addressing label noise by modifying instances to align with noisy labels instead of correcting them, supported by theoretical analysis (§3);
2. We present EchoAlign, a framework featuring EchoMod for controlled instance modification and EchoSelect for strategic sample selection (§4);
3. We empirically validate the benefits of instance modification and demonstrate EchoAlign’s superior performance in noisy environments across three types of noisy data and real-world scenarios, significantly outperforming state-of-the-art methods in accuracy (§5).

2. Related Work

Learning with Noisy Labels Research in this domain has predominantly followed two paths: **(1) Noise-modeling-free methods:** These methods primarily rely on the memorization effects observed in deep neural networks

(DNNs), which tend to learn simpler (clean) examples before memorizing more complex (noisy) ones (Arpit et al., 2017; Wu et al., 2020; Kim et al., 2021). Techniques include early stopping (Han et al., 2018; Nguyen et al., 2020; Liu et al., 2020; Xia et al., 2021; Lu et al., 2022; Bai et al., 2021), pseudo-labeling (Tanaka et al., 2018), and leveraging Gaussian Mixture Models in a semi-supervised learning context (Li et al., 2020). **(2) Noise-modeling methods:** These approaches focus on estimating a noise transition matrix, modeling how clean labels can become corrupted into noisy observations. However, accurately modeling this noise process is particularly challenging when relying solely on noisy data (Xia et al., 2019; Cheng et al., 2020). Many existing studies depend on assumptions that may not hold in real-world datasets (Xia et al., 2020; Yao et al., 2023b; Liu et al., 2023). Consequently, these methods often struggle to effectively handle structured noise patterns, such as subclass-dominant label noise (Bai et al., 2023).

Generative Models Recent advances in generative models, including variational auto-encoders, generative adversarial networks, and diffusion models, have transformed applications with their exceptional sample generation capabilities (Du et al., 2023; Wang et al., 2023; Franceschi et al., 2023). Diffusion models, known for their superior output control, are particularly effective at denoising signals (Zhang et al., 2023; Kingma et al., 2021). While these models hold promise for noisy label scenarios, existing approaches like Dynamics-Enhanced Generative Models (DyGen) (Zhuang et al., 2023) and Label-Retrieval-Augmented Diffusion Models (Chen et al., 2023b) still focus primarily on enhancing predictions or retrieving latent clean labels. Our work takes a fundamentally different approach. We leverage controllable generative models, treating noisy labels as correct and aligning instances with these labels, thus bypassing the challenges of traditional noise modeling and focusing on improving the quality of training data. Controllable generative models, such as ControlNet (Zhang et al., 2023) and iPromptDiff (Chen et al., 2023c), enable precise control over the generated outputs. Unlike traditional generative models which generate images from random noise, controllable generative models use control information (*e.g.*, text descriptions, class labels, or reference images) as input (Bose et al., 2022), guiding the generation process to ensure that the outputs align with the desired characteristics.

3. Analysis

Problem Definition In addressing the challenges posed by learning from noisy labels (LNL), we formally define the problem and introduce the concept of instance modification within a mathematical framework. Let $\mathcal{X}$ represent the input space of instances and $\mathcal{Y}$ the space of labels. In the traditional LNL setting, each instance $X \in \mathcal{X}$ is associated with a noisy label $\tilde{Y} \in \mathcal{Y}$, which may differ from the true label $Y \in \mathcal{Y}$. The goal is to learn a mapping $f : \mathcal{X} \rightarrow \mathcal{Y}$ that predicts the true label Y as accurately as possible, despite the presence of noisy labels. Instance modification diverges from the conventional approach of directly correcting noisy labels $\tilde{Y}$ to match the true labels Y . Instead, we propose adjusting each instance X to better align with its given noisy label $\tilde{Y}$. Mathematically, this involves transforming each instance X into a modified instance X' , such that $f(X')$ aligns more closely with $\tilde{Y}$, leveraging the inherent information contained within the noisy label itself.

Theoretical Analysis According to the causal learning framework (Liu et al., 2023; Yao et al., 2021), the noise can often be represented as a function of both the instance features and external factors, encapsulated by latent variables Z . We assume that the causal relations (commonly occurring in crowdsourcing scenarios) are represented by the causal graph as illustrated in Figure 3, where Z represents latent variables that affect both X and $\tilde{Y}$ indirectly through X . Instance modification aims to transform X into X' such that the modified instance X' better aligns with $\tilde{Y}$ under the assumption that $\tilde{Y}$ contains partial information about the true label Y . Accordingly, we can deduce the effectiveness of instance modification as follows.

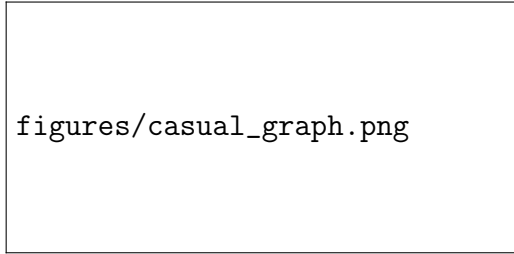


Figure 3: A graphical causal model, revealing a data generative process with instance-dependent label noise.

Theorem 3.1 (Effectiveness of Instance Modification). *Assume that the noisy labels are generated by a stochastic process influenced by latent variables Z ,*

where $\tilde{Y} = h(Y, Z)$ and Y are the true labels. Let T be a transformation such that $X' = T(X, \tilde{Y}; \theta)$, where θ is chosen to optimize the alignment of X' with $\tilde{Y}$. Then, under this transformation, the predictive performance of a model trained on $(X', \tilde{Y})$ is theoretically improved compared to a model trained on $(X, \tilde{Y})$ in terms of:

1. **Alignment:** The mutual information between X' and $\tilde{Y}$, $I(X'; \tilde{Y})$, is maximized relative to $I(X; \tilde{Y})$, indicating better alignment of modified instances with their noisy labels.
2. **Error Reduction:** Compared to a model trained on the original instances X , the expected prediction error $\mathbb{E}_{X', Y}[(Y - f(X'))^2]$ is minimized, where f is the prediction function trained using the modified instances X' . This assumes that the distribution of X' does not deviate significantly from the distribution of X , ensuring that the learned model generalizes well to the original distribution.
3. **Estimation Stability:** The variance of the estimator f is reduced when using X' compared to X , resulting in more stable predictions.
4. **Generalization:** Modifications in X' lead to better generalization. By transforming the original instances to better align with their noisy labels, the model trained on X' is less likely to overfit to the noise and more capable of capturing the true underlying patterns in the data.

This improvement is contingent upon the assumption that the noise model h and the transformation T are appropriately defined and that the latent variable model adequately captures the underlying causal structure of the data. **More details and proofs can be found in Appendix A.**

Theorem 3.1 suggests that instance modification, by aligning more closely with noisy but informative labels $\tilde{Y}$, can leverage the inherent structure and causality in the data to enhance learning. It demonstrates that instance modification improves alignment between instances and noisy labels, reduces information loss, and ultimately leads to better generalization. These insights provide several key motivations for the design of our method. First, the improvements in alignment highlight the importance of modifying instances to embed noisy label information directly. This motivates the use of controllable generative models in EchoAlign, which can effectively incorporate label information into the instance features. Second, ensuring a minimal distribution difference between X and X' is crucial. EchoMod generates X' with small distribution differences from X , while EchoSelect retains clean

samples to control distribution differences, ensuring better generalization on test data. Third, the improvement in estimation stability indicates that using modified features can result in more consistent and reliable model predictions, motivating a focus on preserving the essential characteristics of the data during transformation to reduce variability and enhance both statistical stability and robustness in model performance.

Analyzing Feature Similarity Distributions In this study, we address the challenges of instance modification, which can induce distribution shifts between the training and test sets. Preserving clean original instances is crucial to mitigating these shifts. Existing sample selection methods (*e.g.*, small loss (Han et al., 2018)) often falter under complex label noise conditions, such as instance-dependent noise, necessitating a more precise selection strategy. To this end, we find an interesting phenomenon: Clean samples generally exhibit higher similarity between features of original and modified images, indicating minimal semantic and label changes after modification, whereas noisy samples display lower similarity due to significant semantic and label adjustments. Utilizing the feature similarity distributions between original and modified instances emerges as a robust tool for enhancing sample selection accuracy. These distinctions are visually represented in Figure 4. The similar-

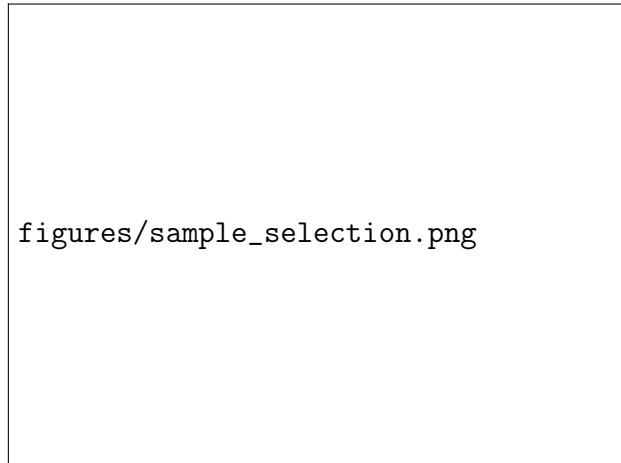


Figure 4: The feature similarity between the original and modified instances is a valuable metric for sample selection after instance modification.

ity is computed using the CLIP ViT-B-32 feature extractor (Radford et al., 2021) on the CIFAR-10 dataset with 30% instance-dependent noise. We use ControlNet (Zhang et al., 2023) to modify instances. The black dashed line

indicates the sample threshold achievable by the previous best method at 96% accuracy (Yang et al., 2022). In contrast, EchoSelect, at 96% accuracy, can retain the samples in the yellow section. In environments with 30% instance-dependent noise, EchoSelect retains nearly twice as many samples at 99% accuracy. Statistical validation using the Kolmogorov-Smirnov test confirmed significant differences in the distributions (p-value < 0.001), demonstrating the utility of feature similarity as a robust metric for identifying clean samples within noisy datasets.

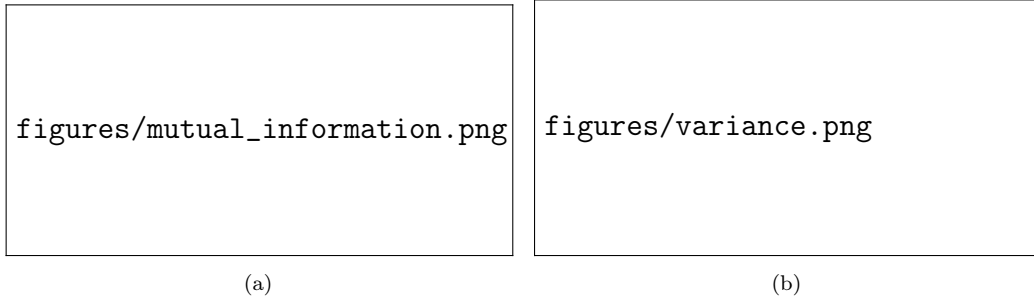


Figure 5: (a) illustrates the mutual information between the labels of 50,000 original samples and their corresponding 50,000 modified samples under 50% instance-dependent noise on CIFAR-10. (b) shows the distribution of the predictive probability of the estimator f using X' and X .

Empirical Validation of Theoretical Analysis To validate the correctness of our proposed Theorem 3.1, we undertook specific experiments to demonstrate its efficacy. The theorem posits that by applying an appropriate transformation T , the alignment between the instances X and the noisy labels $\tilde{Y}$ can be optimized, thereby increasing their mutual information. On the CIFAR-10 dataset, we calculated the mutual information between 50,000 images and their labels. As observed in Figure 5a, the mutual information $I(X'; \tilde{Y})$ between the transformed instances X' and the noisy label $\tilde{Y}$ is significantly higher than the mutual information $I(X; \tilde{Y})$ between the original instances X and $\tilde{Y}$. Figure 5b also supports the third point of our theorem, i.e., the estimator trained on X' has lower variance than the one trained on X , which illustrates the higher stability and robustness of our method. Furthermore, concerning prediction error, Figure 6 displays the training and testing results under different noise types. The results show that using modified samples results in significantly lower errors, both in the training and testing sets.

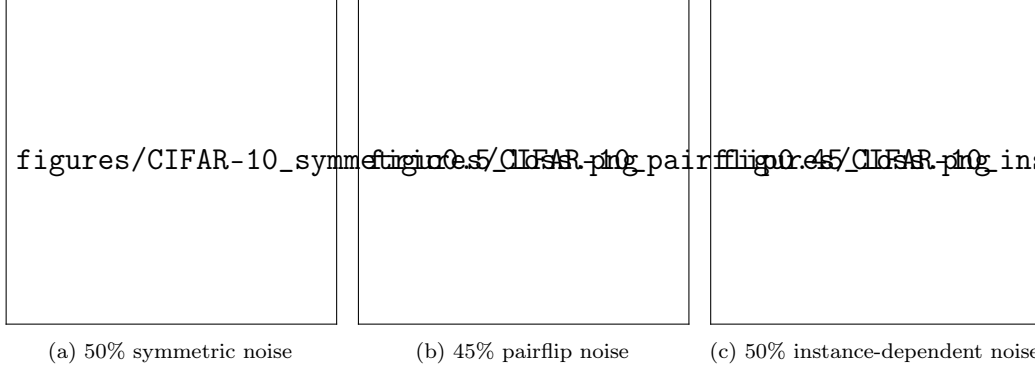


Figure 6: Figures (a), (b), and (c) respectively illustrate the differences in training and testing losses between EchoAlign and the CE model under 50% symmetric noise, 45% pairflip noise, and 50% instance-dependent noise conditions on the CIFAR-10. The **bright peach red** and **deep burgundy** lines represent the performance of CE and EchoAlign on the test set, respectively, while the **light purple** and **light coral pink** lines denote their performance on the training set.

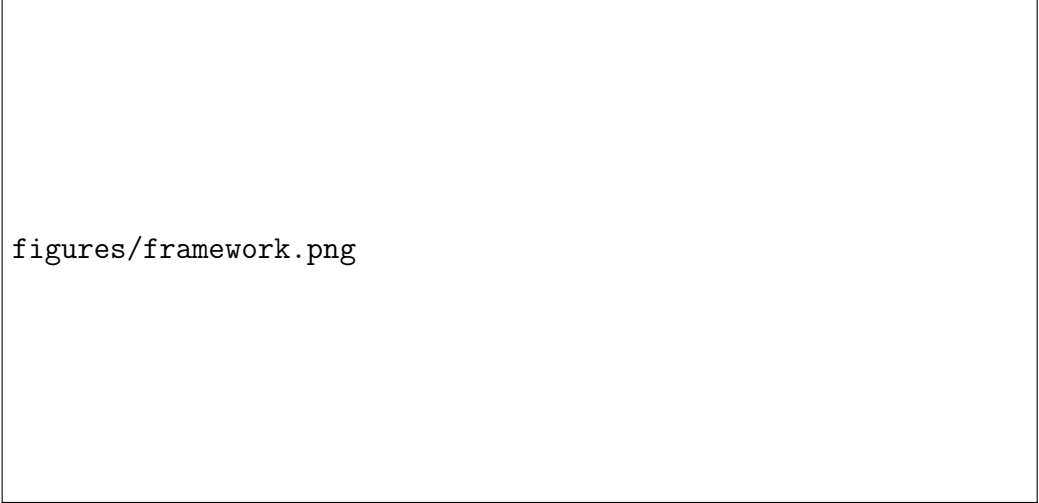
4. EchoAlign

The EchoAlign framework tackles the challenge of noisy labels in supervised learning. It comprises two primary components: **(1)** EchoMod modifies instances using controllable generative models, ensuring alignment with noisy labels while preserving intrinsic characteristics. **(2)** EchoSelect selects original instances with correct labels, maintaining a balanced distribution between original and modified data.

Figure 7 illustrates the overall framework of EchoAlign. Specifically, EchoMod utilizes controllable generative models (CGMs) to perform instance modifications based on noisy labels, while EchoSelect applies feature similarity evaluation to strategically filter instances, preserving original instances that are likely clean and adopting modified instances to ensure label alignment. The integration of these modules effectively addresses the characteristic and distribution shifts introduced by label noise.

4.1. EchoMod: Instance Modification

Motivation When labels are noisy, they do not reflect the true characteristics of the corresponding data instances. This discrepancy hinders a model’s ability to learn meaningful patterns. EchoMod addresses this by transforming data instances to be consistent with their noisy labels. This



figures/framework.png

Figure 7: The framework of EchoAlign.

controlled modification helps the model extract relevant information even when labels contain noise.

Mechanism EchoMod leverages a pre-trained controllable generative model (*e.g.*, a controllable diffusion-based model) to modify data instances. The primary goal is to enhance the alignment between instances and their potentially noisy labels. This alignment is achieved by carefully guiding the generative model’s process. First, the controllable generative model has undergone prior training on a large dataset. This pre-training has equipped the model with a deep understanding of the patterns and structures inherent in the data domain. Second, EchoMod provides both the original instance (X) and the noisy label ($\tilde{Y}$) as inputs to the generative model. This dual conditioning shapes the output, encouraging the model to produce a modified instance (X') that closely aligns with the noisy label while still preserving essential characteristics of the original data. Striking this balance between label alignment and preventing excessive distortion is crucial.

Effectiveness and Flexibility Handling label noise in noisy label learning is a well-recognized challenge. Previous works have primarily focused on the utilization and optimization of internal data. Our approach introduces a novel perspective by integrating external knowledge to enhance model robustness. This integration does not compromise fairness, as our flexible framework accommodates various generative models, and can be fine-tuned for specific noisy label problems. By doing so, we ensure that the method

does not overly rely on any particular model. This approach is particularly advantageous when dealing with ambiguous data. The inherent ambiguity in the data can lead to low confidence in direct discrimination. Instead, by modifying the input data through controllable generative models, we can better resolve discrepancies between instances and labels while preserving the meaningful characteristics of the original data. This not only improves the model’s discriminative capability but also enhances overall performance and reliability.

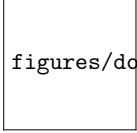
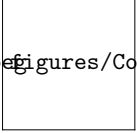
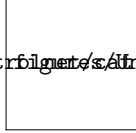
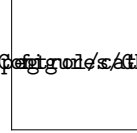
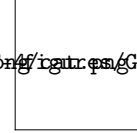
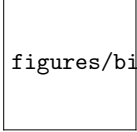
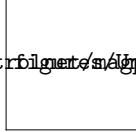
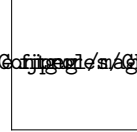
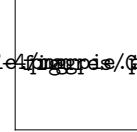
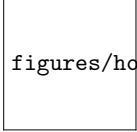
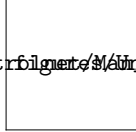
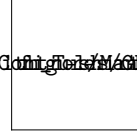
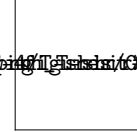
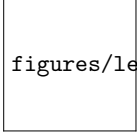
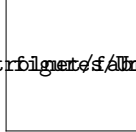
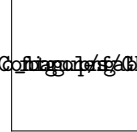
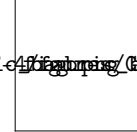
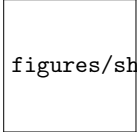
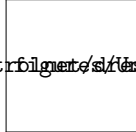
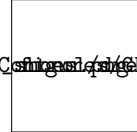
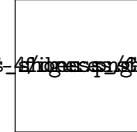
Flexible Generalization with Minimal Tuning In most cases, EchoMod can leverage a pre-trained controllable generative model without fine-tuning during the alignment process. This preserves the model’s ability to understand general data characteristics, promoting EchoAlign’s applicability across various domains. While EchoMod can be effective without fine-tuning, additional performance gains might be realized by tailoring the controllable generative model to highly specialized tasks or data distributions. In such cases, fine-tuning could lead to better alignment between instances and noisy labels, especially in specialized applications such as medical imaging or scientific data.

Visualization and Comparison of Generation Examples To further illustrate the significance of controllable generative models (CGM) and their advantages over non-controllable generative models (NGM), we provide additional generation examples in Table 1. Specifically, we compare the modifications produced by two representative controllable generative models (ControlNet and UniControl) against two advanced non-controllable generative models (GPT-4 and Gemini).

As depicted, CGMs successfully maintain intrinsic instance characteristics while aligning instances effectively with their noisy labels. In contrast, NGMs struggle to preserve crucial features, often leading to semantic misalignments or unrealistic modifications. For instance, when converting a hoodie to a T-shirt, CGMs effectively adjust clothing style while preserving facial and body features, whereas NGMs drastically distort or remove essential details.

These examples empirically validate our theoretical analysis (§ 3), demonstrating that CGMs substantially enhance instance-label alignment, reduce information distortion, and facilitate stable predictions in noisy label scenarios. Thus, controllable generative models emerge as a more robust and reliable choice for instance modification tasks.

Table 1: Results of Controllable and Non-Controllable Generative Models

Noisy Label	Original Instance	CGM		NGM	
		ControlNet	UniControl	GPT-4	Gemini
Cat					
	figures/dog.jpg	figures/ControlNet/cat.jpg	figures/UniControl/cat.jpg	figures/GPT-4/cat.jpg	figures/Gemini/cat.jpg
Magpie					
	figures/bird.jpg	figures/ControlNet/magpie.jpg	figures/UniControl/magpie.jpg	figures/GPT-4/magpie.jpg	figures/Gemini/magpie.jpg
T-shirt					
	figures/hoodie.jpg	figures/ControlNet/T-shirt.jpg	figures/UniControl/T-shirt.jpg	figures/GPT-4/T-shirt.jpg	figures/Gemini/T-shirt.jpg
Fabric bag					
	figures/leather bag.jpg	figures/ControlNet/fabric bag.jpg	figures/UniControl/fabric bag.jpg	figures/GPT-4/fabric bag.jpg	figures/Gemini/fabric bag.jpg
Dress shoes					
	figures/shoes.jpg	figures/ControlNet/dress shoes.jpg	figures/UniControl/dress shoes.jpg	figures/GPT-4/dress shoes.jpg	figures/Gemini/dress shoes.jpg

4.2. EchoSelect: Instance Selection

Motivation While EchoMod improves the alignment between instances and noisy labels, some modified instances may still exhibit inconsistencies. Additionally, the instance modification process can introduce distribution shifts between the modified training data and the true test distribution. EchoSelect safeguards against these issues by identifying and retaining only the most reliable instances after modification. This filtering enhances model robustness, reduces the impact of noisy data, and mitigates distribution shifts introduced by instance modification.

Mechanism EchoSelect employs a metric to assess the similarity between modified instances and a reference representation of clean data. We use the Cosine similarity between feature vectors extracted using a suitable

feature extractor (*e.g.*, the image encoder of CLIP (Radford et al., 2021)):

$$S(X', X) = \frac{\mathbf{z}(X') \cdot \mathbf{z}(X)}{\|\mathbf{z}(X')\| \|\mathbf{z}(X)\|}, \quad (1)$$

where X' and X are modified and original instances, and $\mathbf{z}$ denotes the feature extractor.

Selection Process EchoSelect calculates similarity for all modified instances, comparing them to their original counterparts. To mitigate distribution shifts, priority is given to maintaining clean original instances as much as possible. The final training set consists of two parts: (1) Original instances with similarity above a determined threshold are deemed sufficiently aligned with clean data characteristics and retained, and (2) modified instances with similarity below the threshold are included. These instances are likely those where the modification was most beneficial in aligning them with the noisy labels, while also indicating some degree of difference from the original distribution. The threshold τ balances the inclusion of modified instances with the preservation of the original data distribution, ensuring that only instances aligned with the characteristics of clean data are retained. Our sensitivity analysis (§5.3) confirms the robustness of τ across various types of noise.

4.3. EchoAlign: Optimized Combination

Algorithm 1 EchoAlign Framework

Require: Pre-trained controllable generative model f_θ , Noisy dataset $(X, \tilde{Y})$, Threshold τ , Feature Extractor

Ensure: Refined training dataset

- 1: Generate modified instances: $X' \leftarrow f_\theta(X, \tilde{Y})$
 - 2: Compute similarity: $S(X', X)$ using Equation equation 1
 - 3: # Construct a refined dataset with two parts
 - 4: **Part 1: Original Instances**
 - 5: Select original instances where $S(X', X) \geq \tau$
 - 6: **Part 2: Modified Instances**
 - 7: Select modified instances where $S(X', X) < \tau$
 - 8: Combine Part 1 and Part 2 to form the refined dataset
 - 9: Return the refined dataset
-

The integration of EchoMod and EchoSelect enables the creation of a refined training dataset that is aligned with noisy labels and filtered for

quality. This optimized dataset is better suited for robust learning in the presence of large label noise. Since the refined training dataset can be further used to train a supervised or self-supervised model for LNL, EchoAlign can be combined with advanced LNL methods to further mitigate the impact of label noise. The integration of EchoMod and EchoSelect is encapsulated in Algorithm 1, which details the steps for modifying instances and selecting the optimal subset.

5. Experiments

To evaluate the robustness and effectiveness of our proposed method, we conducted a comprehensive set of experiments across multiple datasets and baseline comparisons. The detailed implementation settings, including model configurations, hyperparameters, and data preprocessing, are provided in Section 5.1.

5.1. Experiment Setup

Dataset Our experiments are conducted on two synthetic datasets: CIFAR-10 and CIFAR-100 (Krizhevsky et al., 2009), and a real-world dataset: CIFAR-10N (Wei et al., 2022). CIFAR-10 and CIFAR-100 each contain 50,000 training and 10,000 testing images, with a size of 32×32 , covering 10 and 100 classes respectively. CIFAR-10N utilizes the same training images from CIFAR-10 but with labels re-annotated by humans. Although CIFAR-10 is a clean dataset, inherent ambiguity in many images leads to prevalent label noise, as even humans struggle to provide consistent labels, a phenomenon reflected in CIFAR-10N. Following previous research protocols (Bai et al., 2021; Xia et al., 2019, 2023b), we corrupted these synthetic datasets using three types of label noise. Specifically, symmetric noise randomly alters a proportion of labels to different classes to simulate random errors; pair flip noise changes labels to adjacent classes with a certain probability; and instance-dependent noise modifies labels based on image features to related incorrect classes. Due to the inherent ambiguity in CIFAR-10N images, correcting label noise has limited impact on performance, making it a more practical choice over Clothing1M (Xiao et al., 2015). A detailed runtime analysis is provided in Section 5.4, demonstrating that the runtime is reasonable across different datasets and can be further optimized using model acceleration techniques. For CIFAR-10N, we use four noisy label sets: ‘Random i=1, 2, 3’, each

representing the label provided by one of three independent annotators; and ‘Worst’, which selects the noisiest label when incorrect annotations are present.

Baseline We compare EchoAlign against various paradigms of baselines for addressing label noise. Under the robust loss function paradigm, we include APL (Ma et al., 2020), PCE (Menon et al., 2019), AUL (Zhou et al., 2023), and CELC (Wei et al., 2023); under the loss correction paradigm, we adopt T-Revision (Xia et al., 2019) and Identifiability (Liu et al., 2023); under the label correction paradigm, we select Joint (Tanaka et al., 2018); and under the sample selection paradigm, we employ Co-teaching (Han et al., 2018), SIGUA (Han et al., 2020), and Co-Dis (Xia et al., 2023a). We compare these methods against a simple cross-entropy (CE) loss baseline. Following the fair baseline design proposed by Xia et al. (2023b), we do not compare with methods such as MixUp (Zhang et al., 2018), DivideMix (Li et al., 2020), and M-correction (Arazo et al., 2019), as these involve semi-supervised learning, making such comparisons unfair due to inconsistent settings.

Implementation Details All experiments were conducted on an NVIDIA V100 GPU using PyTorch. The model architectures and parameter settings were kept consistent with previous studies (Bai et al., 2021). The experiments were configured with a learning rate of 0.1, using the Stochastic Gradient Descent (SGD) optimizer with a momentum of 0.9, and a weight decay set to 1×10^{-4} . We applied 30% and 50% symmetric noise and 45% pair flip noise on the CIFAR-10 and CIFAR-100 datasets to assess model performance. The CIFAR-10 dataset utilized the standard ResNet-18 (He et al., 2016) architecture, while CIFAR-100 used ResNet-34. For the CIFAR-10N dataset, the same ResNet-18 model was used. Prior to training, ControlNet was utilized as our reference model in the controllable generation model module. This choice was strategic; ControlNet was the least effective model identified in prior analyses (Chen et al., 2023c). Employing this model underscores the robustness of our approach, ensuring that the efficacy of our method is not overly contingent upon the capabilities of any specific generative model. This decision highlights our method’s adaptability and general efficacy across varying scenarios. We employed the Canny edge detector as a simple preprocessor to extract features from the instances, using labels as textual controls with the prompt "a photo of {label}". No additional or negative prompts were used, and the sampling process was limited to 20 steps. All experiments were repeated three times with different random seeds, and results are reported as averages with standard deviations.

Data preprocessing For all datasets, including CIFAR-10, CIFAR-

100, and CIFAR-10N, we adopted a unified data augmentation strategy. Specifically, we first applied 4-pixel padding, followed by random cropping to 32×32 pixels. We then applied random horizontal flipping and normalization.

Hyperparameter settings For the ControlNet controllable generation model, we used the simplest Canny preprocessor with both the low threshold and high threshold set to 75. The prompt used was “a photo of label” without any additional prompts or negative prompts. The feature maps output by the preprocessor and the generated images both had a medium size of 512×512 pixels. The diffusion process consisted of 20 steps. For the EchoSelect section, the default threshold was set to 0.4 for all cases with 30% noise, and 0.52 for cases with 45% and 50% noise. The hyperparameters for the training are detailed in Table 2.

Table 2: Training hyperparameters for CIFAR-10/CIFAR-10N and CIFAR-100.

	CIFAR-10/CIFAR-10N	CIFAR-100
architecture	ResNet-18	ResNet-34
optimizer	SGD	SGD
loss function	CE	CE
learning rate(lr)	0.1	0.1
lr decay	100th and 150th	100th and 150th
weight decay	10^{-4}	10^{-4}
momentum	0.9	0.9
batch size	128	128
training samples	45,000	45,000
training epochs	200	200

5.2. Main Results

We evaluated our method on two synthetic datasets (CIFAR-10 and CIFAR-100) and one real-world dataset (CIFAR-10N). For CIFAR-10 and CIFAR-100, 90% of the noisy-labeled data was used for training, 10% for validation, and evaluation was performed on clean test samples. Several baseline results were obtained from previous work (Xia et al., 2023b). As shown in Table 3, our method achieved state-of-the-art performance in most scenarios. Under challenging noise conditions (e.g., 50% instance-dependent noise on CIFAR-10 and 45% symmetric noise on CIFAR-100), our method significantly outperformed existing baselines, demonstrating its robustness against various types of noise. This robustness is particularly attributable to

Table 3: Comparison of test accuracy (%) with state-of-the-art methods on synthetic datasets CIFAR-10 and CIFAR-100. The best three results are bolded and the best one is underlined.

Datasets	Methods	Symmetric		Pairflip	Instance	
		30%	50%	45%	30%	50%
CIFAR-10	CE	73.17 $\pm$ 1.13	52.59 $\pm$ 0.70	51.49 $\pm$ 0.42	71.56 $\pm$ 0.19	49.20 $\pm$ 0.42
	APL	85.54 $\pm$ 0.51	78.36 $\pm$ 0.47	80.84 $\pm$ 0.72	77.57 $\pm$ 0.15	39.45 $\pm$ 6.51
	PCE	86.12 $\pm$ 0.85	74.03 $\pm$ 4.96	65.08 $\pm$ 3.41	85.64 $\pm$ 0.72	64.82 $\pm$ 4.13
	AUL	88.09 $\pm$ 0.78	82.81 $\pm$ 1.16	56.80 $\pm$ 2.69	86.35 $\pm$ 0.90	60.75 $\pm$ 3.77
	CELC	82.51 $\pm$ 0.22	85.08 $\pm$ 3.95	85.72 $\pm$ 4.52	86.67 $\pm$ 1.47	61.85 $\pm$ 4.98
	T-Revision	88.39 $\pm$ 0.38	83.40 $\pm$ 0.65	83.61 $\pm$ 1.06	89.07 $\pm$ 0.35	66.93 $\pm$ 4.14
	Identifiability	87.12 $\pm$ 1.69	83.43 $\pm$ 2.11	83.65 $\pm$ 2.46	80.47 $\pm$ 1.54	55.25 $\pm$ 3.78
	Joint	89.34 $\pm$ 0.52	85.06 $\pm$ 0.29	80.52 $\pm$ 1.90	88.41 $\pm$ 1.02	64.12 $\pm$ 3.89
	Co-teaching	88.93 $\pm$ 0.56	74.02 $\pm$ 0.04	84.19 $\pm$ 0.68	87.07 $\pm$ 0.35	60.09 $\pm$ 3.31
	SIGUA	83.19 $\pm$ 1.26	77.92 $\pm$ 3.11	70.39 $\pm$ 1.94	82.90 $\pm$ 2.00	30.95 $\pm$ 9.70
	Co-Dis	89.20 $\pm$ 0.13	85.36 $\pm$ 0.94	85.02 $\pm$ 1.33	87.13 $\pm$ 0.25	62.77 $\pm$ 3.90
Ours		90.98 $\pm$ 0.20	87.95 $\pm$ 0.12	87.42 $\pm$ 0.11	89.18 $\pm$ 0.20	77.81 $\pm$ 0.30
CIFAR-100	CE	50.99 $\pm$ 1.29	34.5 $\pm$ 0.96	37.03 $\pm$ 0.41	50.33 $\pm$ 2.14	34.70 $\pm$ 1.45
	APL	55.78 $\pm$ 0.91	46.96 $\pm$ 0.81	49.55 $\pm$ 1.05	43.30 $\pm$ 1.57	29.01 $\pm$ 0.09
	PCE	58.84 $\pm$ 1.32	42.63 $\pm$ 2.02	41.05 $\pm$ 2.83	55.72 $\pm$ 1.96	38.72 $\pm$ 3.01
	AUL	69.89 $\pm$ 0.21	60.00 $\pm$ 0.40	39.37 $\pm$ 1.61	67.75 $\pm$ 1.84	40.27 $\pm$ 1.76
	CELC	67.96 $\pm$ 1.88	60.71 $\pm$ 2.39	52.53 $\pm$ 3.17	66.25 $\pm$ 1.93	47.52 $\pm$ 3.93
	T-Revision	62.97 $\pm$ 0.46	43.60 $\pm$ 0.94	49.33 $\pm$ 1.10	56.46 $\pm$ 1.45	40.78 $\pm$ 1.75
	Identifiability	50.53 $\pm$ 1.52	34.87 $\pm$ 2.36	38.16 $\pm$ 2.68	52.48 $\pm$ 1.93	36.72 $\pm$ 3.10
	Joint	63.69 $\pm$ 0.84	55.62 $\pm$ 1.68	49.77 $\pm$ 1.15	64.15 $\pm$ 1.11	45.47 $\pm$ 2.73
	Co-teaching	59.49 $\pm$ 0.36	52.19 $\pm$ 1.42	47.53 $\pm$ 1.39	56.71 $\pm$ 1.26	42.09 $\pm$ 1.73
	SIGUA	54.22 $\pm$ 0.90	50.64 $\pm$ 3.92	39.92 $\pm$ 2.33	53.19 $\pm$ 2.64	38.50 $\pm$ 1.69
	Co-Dis	64.02 $\pm$ 1.37	54.55 $\pm$ 2.06	50.02 $\pm$ 2.80	59.15 $\pm$ 1.92	43.38 $\pm$ 1.25
Ours		68.16 $\pm$ 0.53	60.78 $\pm$ 0.46	60.31 $\pm$ 0.37	65.68 $\pm$ 0.48	57.21 $\pm$ 0.60

EchoMod’s noise-independence, which enables the model to learn consistent features across different noise types and levels. Performance variations were mainly caused by differences in the number of clean samples in the datasets. On the real-world CIFAR-10N dataset, our method also outperformed state-of-the-art methods across all noise settings, exhibiting strong robustness with minimal variations in performance.

5.3. In-Depth Analyses

Ablation Analysis To assess the effectiveness of EchoMod and EchoSelect, we conducted ablation studies by systematically disabling these components. Specifically, we evaluated two configurations: “Instance Modification Only” and “EchoSelect Only,” and compared both against the standard Cross-Entropy Loss (CE) as a baseline. These experiments were carried out under several settings with high noise rates, presenting significant challenges for the model. The experimental results in Table 5 revealed that when using

Table 4: Comparison of test accuracy (%) with state-of-the-art methods on real-world datasets CIFAR-10N. The best three results are bolded and the best one is underlined.

Datasets	Methods	Random 1	Random 2	Random 3	Worst
CIFAR-10N	CE	83.17 $\pm$ 0.48	82.74 $\pm$ 0.42	82.90 $\pm$ 0.28	76.57 $\pm$ 0.23
	APL	84.40 $\pm$ 0.26	84.45 $\pm$ 0.50	84.35 $\pm$ 0.43	78.16 $\pm$ 0.17
	PCE	63.06 $\pm$ 0.37	62.26 $\pm$ 0.36	35.47 $\pm$ 0.36	33.80 $\pm$ 0.33
	AUL	76.26 $\pm$ 0.28	75.24 $\pm$ 0.20	75.48 $\pm$ 0.40	63.61 $\pm$ 1.62
	CELC	83.11 $\pm$ 0.14	83.09 $\pm$ 0.22	82.60 $\pm$ 0.04	73.49 $\pm$ 0.50
	T-Revision	80.99 $\pm$ 0.26	78.99 $\pm$ 1.59	78.80 $\pm$ 1.87	78.37 $\pm$ 0.96
	Identifiability	82.52 $\pm$ 0.87	81.97 $\pm$ 0.85	82.09 $\pm$ 0.73	71.62 $\pm$ 1.16
	Joint	88.20 $\pm$ 0.29	87.54 $\pm$ 0.33	87.67 $\pm$ 0.22	84.29 $\pm$ 0.40
	Co-teaching	82.28 $\pm$ 0.13	82.45 $\pm$ 0.23	82.09 $\pm$ 0.24	79.62 $\pm$ 0.25
	SIGUA	87.67 $\pm$ 1.18	89.01 $\pm$ 0.34	88.40 $\pm$ 0.42	80.65 $\pm$ 1.29
	Co-Dis	80.81 $\pm$ 0.23	80.36 $\pm$ 0.20	80.76 $\pm$ 0.13	78.12 $\pm$ 0.25
	Ours	89.42 $\pm$ 0.12	89.31 $\pm$ 0.06	89.80 $\pm$ 0.25	84.35 $\pm$ 0.09

only Instance Modification, the model’s accuracy did not exceed the baseline CE, and even decreased. This decline primarily stems from the data distribution shift caused by solely using modified instances, adversely affecting the model’s generalization capability. In contrast, using only EchoSelect improved performance but still fell short of the combined EchoAlign approach. This indicates that although EchoSelect significantly reduces the impact of noise, its effectiveness is limited by the number of available samples.

	CIFAR-10 Pairflip-45%	CIFAR-10 IDN-50%	CIFAR-100 Pairflip-45%	CIFAR-100 IDN-50%
CE	51.49	49.20	37.03	34.70
Instance Modification Only	42.77	44.98	15.69	16.36
EchoSelect Only	79.46	65.77	44.24	41.24
Ours	87.42	77.81	60.31	57.21

Table 5: Comparison with state-of-the-art methods on CIFAR-10 and CIFAR-100 in accuracy (%).

Noise rate	BLTM		Ours
	select. acc.	# of selected examples	# of selected examples
IDN-30%	96%	17673 / 50000	26524 / 50000
	99%	10673 / 50000	19010 / 50000
IDN-50%	94%	8029 / 50000	11660 / 50000
	98%	5098 / 50000	6090 / 50000

Table 6: Comparison of sample selection quality under CIFAR-10 instance-dependent noise.

Sensitivity Analysis The performance of EchoSelect is influenced by the threshold value τ , which affects the number and quality of samples

selected from noisy datasets. According to the assumptions of EchoSelect, the optimal threshold should theoretically be around 0.5. To validate the efficacy of our method, we used the state-of-the-art BLTM (Yang et al., 2022) approach as a baseline, with results directly cited from its original publication. As shown in Table 6, EchoSelect was able to select significantly more samples than BLTM under the same accuracy conditions. Particularly, under 30% instance-dependent noise, when the accuracy reached 99%, EchoSelect retained almost twice as many samples as BLTM. Figure 8a clearly

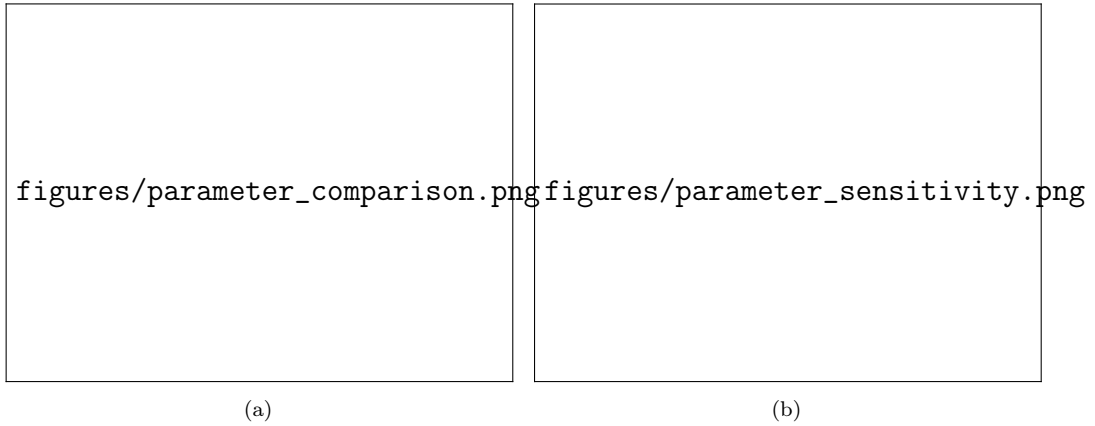


Figure 8: (a) Comparison of the effect of the threshold (τ) on accuracy at different settings of 30% noise rate. (b) Evaluation of thresholding effects on the quality and quantity of sample selection under 30% instance-dependent noise on CIFAR-10.

demonstrates that the threshold τ exhibits robustness and stability across different classes and types of noise, primarily influenced by the noise rate. The slight performance disparity between CIFAR-10 and CIFAR-100 depicted in the figure is attributed to CIFAR-100 containing 20 superclasses, with high similarity among subclasses increasing classification complexity. In practical scenarios, as our method is insensitive to noise types, the noise rate can be efficiently estimated using a validation set (Yu et al., 2018a), even if it is unknown. The optimal threshold can be determined by testing on a simple synthetic dataset. Furthermore, Figure 8b details how adjustments to the threshold value τ affect the quantity and precision of sample selection. The smooth transitions displayed, along with a clearly defined optimal equilibrium region, further affirm the efficacy of our method in various noise environments. To conclude, the threshold τ is robust and insensitive to changes, facilitating practical application.

5.4. Runtime Analysis

Table 7: Comparison of runtime at different settings using an NVIDIA V100-SXM2.

Image resolution	CIFAR-10			Clothing1M
	batch size-1	batch size-8	batch size-16	batch size-16
256×256	31.5	5.5	4.5	129.5
512×512	35.1	18.5	17.2	504.5
768×768	68.5	51.9	/	/

Table 8: Comparison of runtime at different computing performance.

GPU	CIFAR-10	Clothing1M
V100-SXM2	18.5	504.5
RTX 4090	8.5	338.2

The efficiency of EchoMod is significantly influenced by several factors, including the choice of the controllable generative model, the GPU’s floating-point operations per second (FLOPS), the resolution of generated images, batch size, GPU memory capacity, floating-point precision, and the number of diffusion steps if a diffusion model is used. In this study, we employ an NVIDIA V100-SXM2 with 32GB of VRAM, using ControlNet as the benchmark generative model, and apply mixed precision to assess the impacts of image size and batch size on runtime. Runtime is measured in GPU hours, representing the computational time required to perform tasks on a single GPU. As the number of GPUs increases, we observe a super-linear reduction in runtime. Our experiments are conducted on the CIFAR-10 dataset, and we also estimate the runtime for processing the Clothing1M dataset on the same GPU configuration. Table 7 demonstrates that increasing the batch size and reducing the image resolution both significantly impact runtime. We did not conduct tests with image resolution at 768×768 and a batch size of 16 due to GPU memory constraints. Additionally, in Table 8, we compare the effects of different computing performance on runtime. We conducted tests on two different GPUs with an image resolution of 512×512 and a batch size of 8. The NVIDIA V100-SXM2-32GB offers a half-precision compute capability of 125 Tensor TFLOPS and a single-precision capability of 15.7 TFLOPS. In contrast,

the more powerful NVIDIA RTX 4090-24GB GPU provides 165.2 Tensor TFLOPS in half-precision and 82.58 TFLOPS in single-precision. Although using ControlNet as a benchmark for evaluating the Clothing1M dataset is not optimal in terms of inference efficiency, the high flexibility of our framework allows for substantial improvements by incorporating various optimization techniques and more advanced models. For instance, employing optimization strategies from Ultra Fast ControlNet (Paul et al., 2023), such as efficient schedulers and smart CPU memory offloading, can achieve approximately a 70.59% improvement in inference speed. Moreover, adopting more efficient model architectures, such as ControlNet-XS (Zavadski et al., 2024), can significantly enhance inference speed, with reported improvements of up to 46.48%. Based on a preliminary estimation and assuming the use of an RTX 4090-24GB GPU, the inference time on Clothing1M could theoretically be reduced from 338.2 GPU hours to 53.2 GPU hours, even without considering additional memory optimizations that might allow for larger batch sizes.

6. Conclusion

This work provided a novel perspective on treating noisy labels as accurate through instance modification. Theoretical analysis supports that this alignment process allows models to learn meaningful patterns despite the presence of labeling errors. To address the challenges of instance modification, we proposed the EchoAlign framework, which integrates a controllable generative model with strategic sample selection to create a robust training dataset. Extensive experiments on diverse datasets demonstrate the superiority of EchoAlign over existing methods, particularly in scenarios with high levels of label noise. However, EchoAlign’s success partially depends on the capabilities of the controllable generative model, and the cost of fine-tuning these models to adapt to out-of-distribution data could be a barrier in resource-limited settings. Future directions include investigating supervised or self-supervised extensions and broader applications such as medical imaging or real-time systems.

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Appendix A. Proof of the Theorem on the Effectiveness of Instance Modification

To provide a comprehensive proof of the theorem regarding the effectiveness of instance modification in learning from noisy labels, we will assume the definitions and setup described in the theorem’s statement. We will address each component of the theorem, demonstrating how the instance modification approach theoretically leads to improvements in alignment, error reduction, estimation stability, and generalization.

Proof. We prove each component of Theorem 3.1 regarding the effectiveness of instance modification as follows:

1. Alignment:

Claim: The mutual information between X' and $\tilde{Y}$, $I(X'; \tilde{Y})$, is maximized relative to $I(X; \tilde{Y})$, indicating better alignment of modified instances with their noisy labels.

Definitions and Assumptions:

Let $X \in \mathbb{R}^d$ be the original instances with distribution P_X and $\tilde{Y} \in \mathcal{Y}$ be the noisy labels, where $\mathcal{Y}$ is the label space. The modified instances are defined as $X' = T(X, \tilde{Y}; \theta) \in \mathbb{R}^d$, where T is a transformation function parameterized by θ , designed to improve alignment between X' and $\tilde{Y}$.

We make the following assumptions:

- **A1. Transformation Improves Alignment:** The transformation T reduces the conditional entropy of $\tilde{Y}$ given the features:

$$H(\tilde{Y} | X') \leq H(\tilde{Y} | X).$$

This means that the uncertainty in $\tilde{Y}$ given X' is less than or equal to the uncertainty given X .

Goal:

Our aim is to prove that the mutual information between X' and $\tilde{Y}$ is greater than or equal to that between X and $\tilde{Y}$:

$$I(X'; \tilde{Y}) \geq I(X; \tilde{Y}),$$

where the mutual information is defined as:

$$I(X; \tilde{Y}) = H(\tilde{Y}) - H(\tilde{Y} | X).$$

Proof. We begin by expressing the mutual information between X (or X') and $\tilde{Y}$:

$$I(X; \tilde{Y}) = H(\tilde{Y}) - H(\tilde{Y} | X), \quad I(X'; \tilde{Y}) = H(\tilde{Y}) - H(\tilde{Y} | X').$$

The difference in mutual information is then:

$$\begin{aligned} \Delta I &= I(X'; \tilde{Y}) - I(X; \tilde{Y}) \\ &= \left[H(\tilde{Y}) - H(\tilde{Y} | X') \right] - \left[H(\tilde{Y}) - H(\tilde{Y} | X) \right] \\ &= H(\tilde{Y} | X) - H(\tilde{Y} | X'). \end{aligned}$$

According to Assumption **A1**, the transformation T reduces the conditional entropy of $\tilde{Y}$ given the features, so:

$$H(\tilde{Y} | X') \leq H(\tilde{Y} | X).$$

Therefore, the difference ΔI is non-negative:

$$\Delta I = H(\tilde{Y} | X) - H(\tilde{Y} | X') \geq 0.$$

This implies that:

$$I(X'; \tilde{Y}) \geq I(X; \tilde{Y}).$$

Thus, the mutual information between the modified instances X' and the noisy labels $\tilde{Y}$ is greater than or equal to that between the original instances X and $\tilde{Y}$, indicating improved alignment between X' and $\tilde{Y}$. □

2. Error Reduction:

Claim: Compared to a model trained on the original instances X , the expected prediction error $\mathbb{E}_{X', Y}[(Y - f(X'))^2]$ is minimized, where f is the prediction function trained using the modified instances X' . This assumes that the distribution of X' does not deviate significantly from the distribution of X , ensuring that the learned model generalizes well to the original distribution.

Definitions and Assumptions:

Let $X \in \mathbb{R}^d$ be the original instances with distribution P_X , $\tilde{Y} \in \mathbb{R}$ be the noisy labels, and $Y \in \mathbb{R}$ be the true labels. The modified instances are defined as $X' = T(X, \tilde{Y}; \theta) \in \mathbb{R}^d$, where T is a transformation designed to improve alignment between X' and $\tilde{Y}$ while preserving essential predictive information about Y .

We consider two models:

- $f_X : \mathbb{R}^d \rightarrow \mathbb{R}$, trained on $(X, \tilde{Y})$.
- $f_{X'} : \mathbb{R}^d \rightarrow \mathbb{R}$, trained on $(X', \tilde{Y})$.

The loss function $L : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ is assumed to be convex and differentiable with respect to its second argument (e.g., squared loss $L(y, \hat{y}) = (y - \hat{y})^2$).

We make the following assumptions:

- **A1. Transformation Improves Alignment:** The transformation T reduces the variance of the noisy labels conditioned on the features:

$$\text{Var}(\tilde{Y} \mid X') \leq \text{Var}(\tilde{Y} \mid X).$$

- **A2. Transformation Preserves Predictive Information:** The transformation T preserves the essential information needed to predict Y :

$$\text{Var}(Y \mid X') \approx \text{Var}(Y \mid X).$$

Goal:

Our aim is to prove that the expected risk of $f_{X'}$ evaluated on the original instances X is less than or equal to that of f_X :

$$R(f_{X'}) \leq R(f_X),$$

where the expected risks are defined as:

$$R(f_X) = \mathbb{E}_{X,Y} [L(Y, f_X(X))], \quad R(f_{X'}) = \mathbb{E}_{X,Y} [L(Y, f_{X'}(X))].$$

Proof. We decompose the expected risk into the Bayes risk and the excess risk. Let $f^*(X) = \mathbb{E}[Y \mid X]$ be the Bayes optimal predictor, which minimizes the expected loss:

$$R^* = \mathbb{E}_{X,Y} [L(Y, f^*(X))].$$

The excess risks for f_X and $f_{X'}$ are then:

$$\mathcal{E}(f_X) = R(f_X) - R^*, \quad \mathcal{E}(f_{X'}) = R(f_{X'}) - R^*.$$

Our goal is to show that $\mathcal{E}(f_{X'}) \leq \mathcal{E}(f_X)$.

For each model, the excess risk can be expressed as:

$$\mathcal{E}(f) = \mathbb{E}_X [\mathbb{E}_{Y|X} [L(Y, f(X)) - L(Y, f^*(X))]].$$

Assuming L is twice differentiable, we perform a Taylor expansion of $L(Y, f(X))$ around $f^*(X)$:

$$\begin{aligned} L(Y, f(X)) &\approx L(Y, f^*(X)) + L^{(1)}(Y, f^*(X))\delta(X) \\ &\quad + \frac{1}{2}L^{(2)}(Y, f^*(X))\delta(X)^2. \end{aligned}$$

where $\delta(X) = f(X) - f^*(X)$, and $L^{(1)}$, $L^{(2)}$ are the first and second derivatives with respect to the second argument. Since $f^*(X)$ minimizes $\mathbb{E}_{Y|X}[L(Y, f(X))]$, the expected first derivative term is zero:

$$\mathbb{E}_{Y|X}[L^{(1)}(Y, f^*(X))] = 0.$$

Thus, the excess risk simplifies to:

$$\mathcal{E}(f) \approx \mathbb{E}_X \left[\frac{1}{2} \mathbb{E}_{Y|X}[L^{(2)}(Y, f^*(X))] \delta(X)^2 \right].$$

We focus on comparing $\delta_X(X)^2$ and $\delta_{X'}(X)^2$, where $\delta_X(X) = f_X(X) - f^*(X)$ and $\delta_{X'}(X) = f_{X'}(X) - f^*(X)$.

Under Assumption **A1**, the variance of the estimation error is reduced when training on $(X', \tilde{Y})$:

$$\text{Var}(\delta_{X'}(X)) \leq \text{Var}(\delta_X(X)).$$

Assuming the biases $\mathbb{E}_X[\delta_X(X)]$ and $\mathbb{E}_X[\delta_{X'}(X)]$ are negligible or similar due to Assumption **A2**, we have:

$$\begin{aligned} \mathbb{E}_X[\delta_{X'}(X)^2] &= \text{Var}(\delta_{X'}(X)) + (\mathbb{E}_X[\delta_{X'}(X)])^2 \\ &\leq \text{Var}(\delta_X(X)) + (\mathbb{E}_X[\delta_X(X)])^2 = \mathbb{E}_X[\delta_X(X)^2]. \end{aligned}$$

Since $L^{(2)}$ is positive due to the convexity of L , it follows that:

$$\mathcal{E}(f_{X'}) \leq \mathcal{E}(f_X).$$

Adding back the Bayes risk R^* , we conclude:

$$R(f_{X'}) = R^* + \mathcal{E}(f_{X'}) \leq R^* + \mathcal{E}(f_X) = R(f_X).$$

Therefore, the expected prediction error is minimized when using the model trained on the modified instances X' , even when evaluated on the original data X . $\square$

3. Estimation Stability:

Claim: The variance of the estimator f is reduced when using X' compared to X , resulting in more stable predictions.

Formulation: Assume the following linear regression models for simplicity, though the concepts generalize to non-linear models:

- Model using X : $f_X = \beta_X^T X + \epsilon_X$, where ϵ_X is the noise term.
- Model using X' : $f_{X'} = \beta_{X'}^T X' + \epsilon_{X'}$, where $\epsilon_{X'}$ is the noise term for the modified model.

Goal: To demonstrate that the variance of the estimator $f_{X'}$ is lower than that of f_X .

Proof.

- **Model Definitions and Assumptions:** Assume that both β_X and $\beta_{X'}$ are obtained by ordinary least squares (OLS), implying that they minimize the respective mean squared errors. The variance of the estimator in OLS is inversely proportional to the Fisher information of the model, Fisher information matrix $I(\beta)$ is represented as $X^T X$ and $X'^T X'$, reflecting the variability of input features.
- **Variance of Estimators:** The covariance of the estimated coefficients under OLS can be expressed as:

$$\begin{aligned}\text{Cov}(\hat{\beta}_X) &= \sigma^2 (X^T X)^{-1} \\ \text{Cov}(\hat{\beta}_{X'}) &= \sigma^2 (X'^T X')^{-1}\end{aligned}$$

where σ^2 is the variance of the error terms ϵ_X and $\epsilon_{X'}$, assumed equal for simplicity. The variance of the predicted values at any input x and its modified version x' is:

$$\begin{aligned}\text{Var}(f_X(x)) &= x^T \text{Cov}(\hat{\beta}_X) x = \sigma^2 x^T (X^T X)^{-1} x \\ \text{Var}(f_{X'}(x')) &= x'^T \text{Cov}(\hat{\beta}_{X'}) x' = \sigma^2 x'^T (X'^T X')^{-1} x'\end{aligned}$$

- **Comparative Analysis of Variance:** Since X' is designed to be more informative and aligned with $\tilde{Y}$, it is reasonable to assume that X' exhibits higher effective variability in the dimensions that are most relevant for predicting Y . This increased effective variability implies

that the matrix $X'^T X'$ is larger than $X^T X$ in the Loewner partial ordering, meaning:

$$X'^T X' \succeq X^T X$$

This leads to:

$$(X'^T X')^{-1} \preceq (X^T X)^{-1}$$

Because the inverse of a larger positive definite matrix is smaller in the Loewner ordering. Consequently, the covariance matrices satisfy:

$$\text{Cov}(\hat{\beta}_{X'}) \preceq \text{Cov}(\hat{\beta}_X)$$

Assuming that the transformation from x to x' does not significantly increase the norm of the input vectors (i.e., $\|x'\| \approx \|x\|$), we can compare the variances of the predictions:

$$\begin{aligned} \text{Var}(f_{X'}(x')) &= x'^T \text{Cov}(\hat{\beta}_{X'}) x' \leq x'^T \text{Cov}(\hat{\beta}_X) x' \\ &\approx x^T \text{Cov}(\hat{\beta}_X) x = \text{Var}(f_X(x)) \end{aligned}$$

Thus:

$$\text{Var}(f_{X'}(x')) \leq \text{Var}(f_X(x))$$

- **Estimation Stability:** Thus, the variance of the predictions using X' is less than or equal to that using X . The reduction in variance implies that $f_{X'}$ offers more stable and reliable predictions compared to f_X . This stability is crucial when the model is applied in practice, particularly in the presence of noisy data conditions. This result holds under the assumptions that:

1. $X'^T X' \succeq X^T X$ (i.e., $X'^T X' - X^T X$ is positive semidefinite).
2. The transformation from x to x' does not significantly increase the input vector norms.

This detailed proof shows that by focusing on feature dimensions that are more predictive of Y , instance modification via X' not only improves the alignment with the noisy labels but also enhances the stability of the model's predictions.

□

4. Generalization:

Claim: Modifications in X' lead to better generalization. By transforming the original instances to better align with their noisy labels, the model trained on X' is less to overfit to the noise and more capable of capturing the true underlying patterns in the data.

Setup: Let

- X denote the original feature space and $X' = T(X, \tilde{Y}; \theta)$ denote the modified feature space, where T is a transformation (assumed to be Lipschitz continuous with Lipschitz constant $L \leq 1$), and θ is fixed, that optimizes some aspect of the data to better align with noisy labels $\tilde{Y}$. In our EchoAlign framework, the transformation T is implemented using controllable generative models. These models can be designed to be Lipschitz continuous by incorporating techniques like spectral normalization or gradient penalties. Ensuring that $L \leq 1$ is reasonable because we aim for T to be non-expansive, preventing the amplification of noise and promoting stability in the transformation.
- $\mathcal{F}$ be the class of functions $f : \mathcal{X} \rightarrow \mathbb{R}$ considered by the learning algorithm, where $\mathcal{X}$ is either the space of X or X' .

Rademacher Complexity: Rademacher complexity measures the ability of a function class to fit random noise. The Rademacher complexity for the class of functions $\mathcal{F}$ applied to the original features X and the modified features X' are defined respectively as:

$$\mathfrak{R}_n(\mathcal{F}_X) = \mathbb{E}_{\sigma, X} \left[\sup_{f \in \mathcal{F}_X} \frac{1}{n} \sum_{i=1}^n \sigma_i f(X_i) \right]$$

$$\mathfrak{R}_n(\mathcal{F}_{X'}) = \mathbb{E}_{\sigma, X'} \left[\sup_{f \in \mathcal{F}_{X'}} \frac{1}{n} \sum_{i=1}^n \sigma_i f(X'_i) \right]$$

Generalization Bounds: Using these definitions, the generalization bounds for a Lipschitz continuous loss function l can be expressed for both feature sets. Assuming the same hypothesis class $\mathcal{F}$, the bounds are:

$$\mathbb{E}[l(f(X), Y)] \leq \frac{1}{n} \sum_{i=1}^n l(f(X_i), Y_i) + 2\mathfrak{R}_n(\mathcal{F}_X) + c$$

$$\mathbb{E}[l(f(X'), Y)] \leq \frac{1}{n} \sum_{i=1}^n l(f(X'_i), Y_i) + 2\mathfrak{R}_n(\mathcal{F}_{X'}) + c$$

where c is a constant that depends on the complexity of the loss function.

Impact of Instance Modification on Feature Space: The transformation T is designed to adjust features in X to more effectively align with $\tilde{Y}$, potentially reducing the variability of X that is irrelevant to predicting Y . This transformation can:

- Increase the signal-to-noise ratio in X' compared to X .
- Focus the variability in X' on aspects that are more predictive of Y , based on the information contained in $\tilde{Y}$.

Proof. To show that $\mathfrak{R}_n(\mathcal{F}_{X'}) \leq L \cdot \mathfrak{R}_n(\mathcal{F}_X)$ and hence that $\mathfrak{R}_n(\mathcal{F}_{X'}) \leq \mathfrak{R}_n(\mathcal{F}_X)$ when $L \leq 1$, we analyze how the transformation T affects the ability of the function class $\mathcal{F}$ to fit random noise.

Since T is Lipschitz continuous with Lipschitz constant $L \leq 1$, we can apply the contraction principle (Ledoux-Talagrand contraction inequality) to relate the Rademacher complexities:

$$\mathfrak{R}_n(\mathcal{F}_{X'}) = \mathfrak{R}_n(\mathcal{F} \circ T) \leq L \cdot \mathfrak{R}_n(\mathcal{F}_X)$$

Since $X' = T(X, \tilde{Y}; \theta)$ and T is Lipschitz continuous in X (with $\tilde{Y}$ and θ fixed during transformation), the inequality applies directly. When $L \leq 1$, this inequality implies that the Rademacher complexity on the modified data X' is less than or equal to that on the original data X :

$$\mathfrak{R}_n(\mathcal{F}_{X'}) \leq \mathfrak{R}_n(\mathcal{F}_X)$$

Since the generalization error bound depends on the Rademacher complexity, a lower Rademacher complexity implies a tighter generalization bound. Specifically:

$$\mathbb{E}[l(f(X'), Y)] \leq \frac{1}{n} \sum_{i=1}^n l(f(X'_i), Y_i) + 2\mathfrak{R}_n(\mathcal{F}_{X'}) + c$$

With $\mathfrak{R}_n(\mathcal{F}_{X'}) \leq \mathfrak{R}_n(\mathcal{F}_X)$, the bound on the expected loss for X' is tighter than that for X . Therefore, the model trained on X' is expected to generalize better than the model trained on X .

This inequality derived from comparing the Rademacher complexities and the corresponding generalization bounds provides a theoretical basis for asserting that instance modification enhances the model's ability to generalize.

By ensuring that the transformation T is Lipschitz continuous with $L \leq 1$, we have formally shown that the Rademacher complexity decreases or remains the same, leading to improved generalization performance. This proof underscores the importance of feature alignment and relevance in improving machine learning model performance in noisy settings. $\square$

Proof of the Lipschitz Continuity of the Transformation T with $L \leq 1$:

In our framework, the transformation T modifies an original instance X to a modified instance $X' = T(X, \tilde{Y}; \theta)$, aiming to align X' more closely with its noisy label $\tilde{Y}$ while preserving essential characteristics of X . To demonstrate that T is Lipschitz continuous with Lipschitz constant $L \leq 1$, we proceed as follows.

We define the transformation T as a convex combination of the original instance X and an adjustment function $\phi(X, \tilde{Y}; \theta)$ that incorporates the influence of the noisy label:

$$T(X, \tilde{Y}; \theta) = (1 - \alpha)X + \alpha\phi(X, \tilde{Y}; \theta),$$

where $\alpha \in [0, 1]$ is a parameter controlling the degree of modification, and $\phi(X, \tilde{Y}; \theta)$ is designed to adjust X based on $\tilde{Y}$.

To ensure that T is Lipschitz continuous with $L \leq 1$, we require that ϕ itself is Lipschitz continuous with Lipschitz constant $L_\phi \leq 1$. Under this condition, for any two instances $X_1, X_2 \in \mathcal{X}$, we have:

$$\begin{aligned} & \|T(X_1, \tilde{Y}; \theta) - T(X_2, \tilde{Y}; \theta)\| \\ &= \left\| (1 - \alpha)(X_1 - X_2) + \alpha \left(\phi(X_1, \tilde{Y}; \theta) - \phi(X_2, \tilde{Y}; \theta) \right) \right\| \\ &\leq (1 - \alpha)\|X_1 - X_2\| + \alpha\|\phi(X_1, \tilde{Y}; \theta) - \phi(X_2, \tilde{Y}; \theta)\| \\ &\leq (1 - \alpha)\|X_1 - X_2\| + \alpha L_\phi \|X_1 - X_2\| \\ &= ((1 - \alpha) + \alpha L_\phi) \|X_1 - X_2\|. \end{aligned}$$

Since $L_\phi \leq 1$ and $\alpha \in [0, 1]$, we have:

$$(1 - \alpha) + \alpha L_\phi \leq (1 - \alpha) + \alpha = 1,$$

which means that the Lipschitz constant L of T satisfies $L \leq 1$.

To ensure that ϕ has Lipschitz constant $L_\phi \leq 1$, we can design ϕ using various techniques:

- **Spectral Normalization:** Spectral normalization constrains the spectral norm (largest singular value) of each linear layer in the neural network implementing ϕ to be at most 1 (Miyato et al., 2018). By normalizing the weight matrices W of the layers such that:

$$\|W\|_2 = \sigma_{\max}(W) = 1,$$

we ensure that the Lipschitz constant of each layer does not exceed 1. Since the Lipschitz constant of a composition of functions is bounded by the product of the individual Lipschitz constants, and each layer has $L_i \leq 1$, the overall Lipschitz constant of ϕ satisfies $L_\phi \leq 1$.

- **Gradient Penalties:** Incorporating gradient penalties into the training of ϕ encourages the network to have controlled Lipschitz continuity (Gulrajani et al., 2017). We add a regularization term to the loss function:

$$\mathcal{L}_{\text{GP}} = \lambda, \mathbb{E}_{X, \tilde{Y}} \left[\left(\left\| \nabla_X \phi(X, \tilde{Y}; \theta) \right\|_2 - 1 \right)^2 \right],$$

where $\lambda > 0$ is a penalty coefficient. Minimizing $\mathcal{L}_{\text{GP}}$ enforces the gradient norms of ϕ to be close to 1, ensuring $L_\phi \leq 1$.

- **Contractive Autoencoders:** Designing ϕ as a contractive autoencoder (Rifai et al., 2011) involves adding a contraction penalty to the loss function:

$$\mathcal{L}_{\text{CAE}} = \mathbb{E}_X \left[\|X - \phi(X, \tilde{Y}; \theta)\|_2^2 + \lambda \left\| \frac{\partial \phi(X, \tilde{Y}; \theta)}{\partial X} \right\|_F^2 \right],$$

where $\|\cdot\|_F$ denotes the Frobenius norm, and $\lambda > 0$ controls the penalty strength. This penalizes large derivatives, encouraging ϕ to be contractive and thus Lipschitz continuous with $L_\phi \leq 1$.

□

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